

FYS-STK4155 Project 2 Report

Regression and Classification Using a Feedforward Neural Network

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Abstract

This project explores the implementation and performance of a fully connected feedforward neural network (FFNN) applied to both regression and classification problems. The regression task focuses on approximating the one-dimensional Runge function, a well-known benchmark for testing overfitting tendencies in numerical interpolation methods [1]. Accurately fitting this function is challenging due to its sharp curvature near the boundaries, making it a useful case for evaluating how neural networks generalize compared to linear and regularized regression techniques. The classification task involves the multiclass MNIST dataset of handwritten digits [3], serving as a standard benchmark for evaluating neural network performance on high-dimensional data [2].

Our primary objectives are to (i) develop and implement an FFNN from scratch using stochastic gradient descent and adaptive optimizers such as RMSprop and ADAM [1], (ii) compare its performance with that of traditional regression methods from Project 1 (OLS, Ridge, and Lasso), and (iii) investigate the influence of activation functions (Sigmoid, ReLU, and Leaky ReLU), network depth, and regularization on learning performance.

1 Introduction

Machine learning methods have revolutionized how we model and interpret complex relationships in data. While classical regression techniques such as Ordinary Least Squares (OLS), Ridge, and Lasso regression provide interpretable and efficient approaches for many problems, their representational capacity is fundamentally limited to linear or polynomial relationships. Neural networks, in contrast, offer a flexible framework capable of approximating highly nonlinear functions, given sufficient network depth and training data [1].

In this project, we extend the regression framework developed in Project 1 by implementing a feedforward neural network (FFNN) from first principles. The first part of the project focuses on a regression problem based on the one-dimensional Runge function,

$$f(x) = \frac{1}{1 + 25x^2},$$

which is a classical test case in numerical analysis due to its oscillatory behavior and sensitivity to overfitting. The goal is to evaluate how a neural network trained using mean squared error (MSE) compares to traditional regression approaches in terms of predictive accuracy, bias, and stability. We further explore how different optimization methods (stochastic gradient descent, RMSprop, and ADAM) [1], learning rates, and activation functions influence convergence and generalization.

The second part of the project extends the same neural network framework to a multiclass classification task using the MNIST dataset [3]. This dataset of handwritten digits represents a high-dimensional nonlinear classification problem that allows us to test the FFNN's ability to learn complex decision boundaries. Using the cross-entropy loss and Softmax output activation, we compare the network's accuracy against established machine learning libraries such as Scikit-learn or Pytorch [2].

The overall objectives of the project are: (1) to construct a general neural network framework capable of handling both regression and classification, (2) to investigate how architectural and hyperparameter choices affect performance, and (3) to critically compare the neural network's performance with established regression and classification methods. Through these experiments, we aim to illustrate both the strengths and limitations of neural networks relative to more traditional machine learning techniques [1, 2].

2 Week 44 exercises

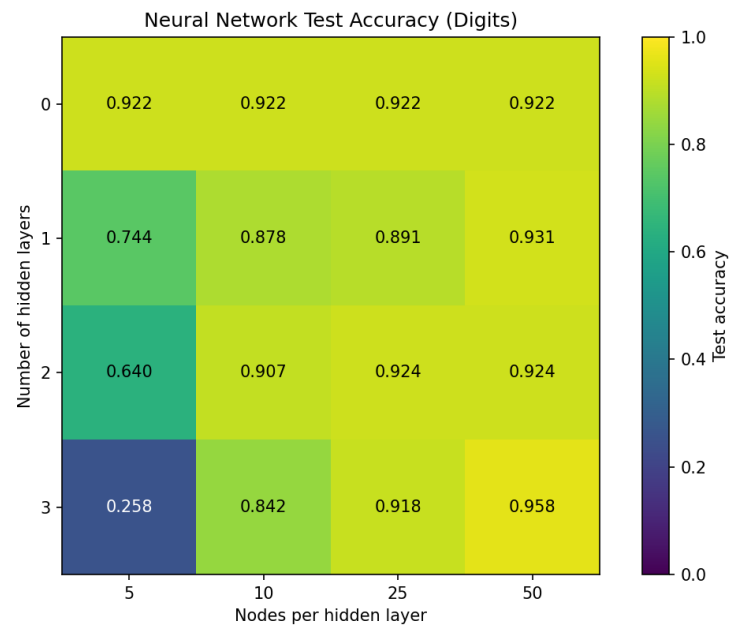


Figure 1: Heatmap showing test accuracy of a neural network with varying numbers of hidden layers [0, 1, 2, 3] and nodes per layer [5, 10, 25, 50] for the MNIST classification task.

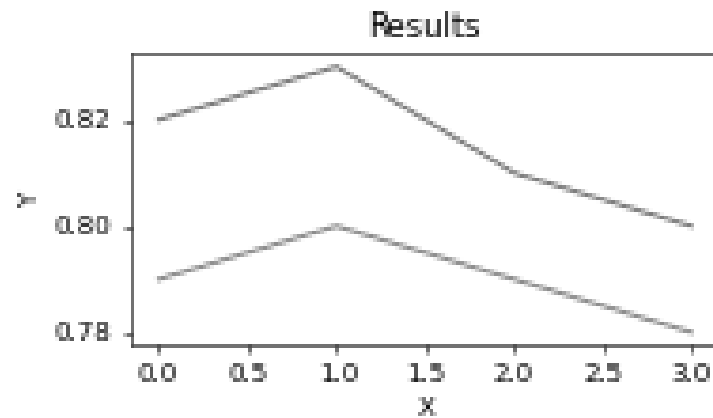


Figure 2: Results

Notes on why this is a poor figure (intentional, not for Project 2 final report): The title ("Results") is vague; it does not say what the result is about. Axes are labeled "X" and "Y" with no units or definitions; the reader cannot tell what is plotted. No legend; multiple lines cannot be attributed to specific models/settings. Poor contrast and nearly identical gray tones. Very small size and low resolution, making text unreadable when embedded. No context: dataset, preprocessing, train/test split, metric definition, and sample size are unstated. Tick marks and ranges are default and uninformative; scale and key values are unclear. The caption text is minimal and uninformative; it does not convey a takeaway or key message.

References

- [1] M. P. Deisenroth, A. A. Faisal, and C. S. Ong, *Mathematics for Machine Learning*. Cambridge University Press, 2020.
- [2] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*. Available: <https://github.com/CompPhysics/MLERasmus/blob/master/doc/Textbooks/elementsstat.pdf>, 2025.
- [3] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.